Indian Institute of Technology Madras
DA3409 Deep Learning - Quiz 2
Time: 40 minutes Maximum marks: 20

Answer all questions. Show your working for numerical questions.
For Q1 and Q2 write only the letter of the correct option.

Q1. Which of the following does NOT reduce the vanishing gradient problem? [1 mark]
(a) ReLU activations (b) Residual connections
(c) Sigmoid activations in every layer (d) Batch normalization

Q2. A convolution layer has 3x3 kernels, 64 input channels and 128 output channels, with one bias per output channel. How many trainable parameters does it have? [1 mark]
(a) 73,728 (b) 73,856 (c) 1,152 (d) 147,584

Q3. A 64x64x3 image is passed through a convolution layer with 16 filters of size 5x5, stride 2 and padding 2. Find the shape of the output and the number of trainable parameters of the layer (including biases). Show your working. [3 marks]

Q4. The output of the layer in Q3 is flattened and fed to a fully connected layer with 10 outputs. How many trainable parameters does this fully connected layer have? Show your working. [3 marks]

Q5. A single neuron computes $y = \text{sigmoid}(wx + b)$ and the loss is $L = 1/2\ (y - t)^2$. Derive dL/dw. [4 marks]

Q6. Explain dropout: what it does during training, what changes at test time, and why it reduces overfitting. [4 marks]

Q7. Write a Python function softmax(x) that takes a list of numbers and returns their softmax probabilities. It must be numerically stable. In one sentence, explain why your approach does not change the result. [4 marks]